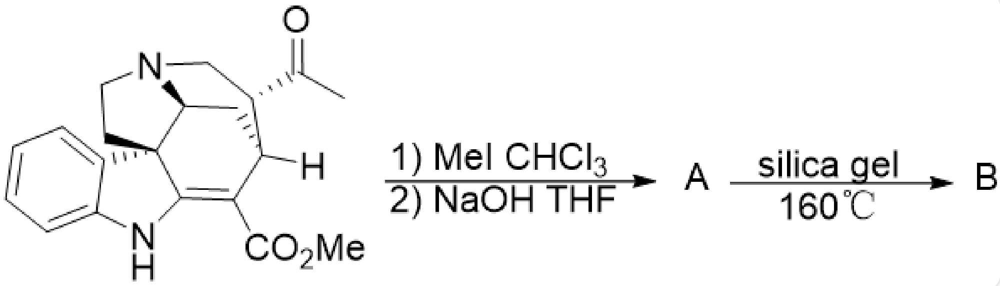
Question

Traditional C-H bond activation involves inducing a transition metal to perform oxidative addition on a C-H bond via an inductive group. However, removing the H atom as a hydride to obtain a positive charge center is also a long-standing and very practical activation strategy. Some researchers have used this approach to complete the reconstruction of natural product skeletons.

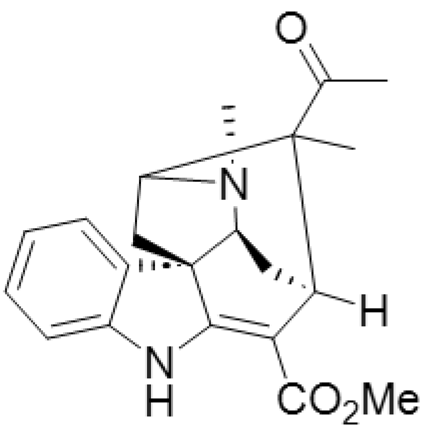


CC([C@H]1CN2[C@H]3C[C@]1([H])C(C(OC)=O)=C4NC5=CC=CC=C5[C@@]43CC2)=O first reacts in $CHCl_3$ and MeI, and then is treated with $NaOH$ in THF solution to obtain $\mathbf{A}$. $\mathbf{A}$ is added to silica gel and can be converted to $\mathbf{B}$ at 160℃.

Give a reasonable structure for product $\mathbf{B}$.

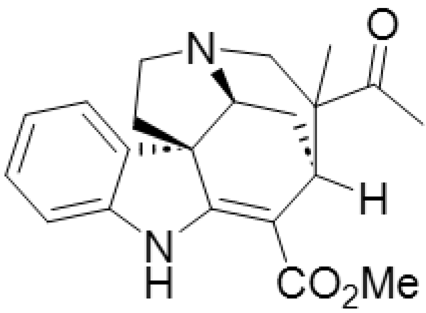
A. All other options are incorrect

B.



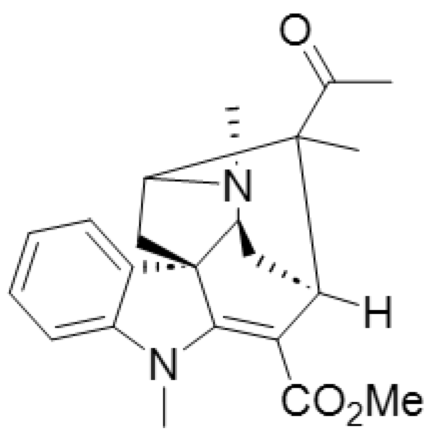
CC1(C(C)=O)[C@@]2([H])C(C(OC)=O)=C3NC4=CC=CC=C4[C@]3(CC1N5C)[C@@H]5C2

C.



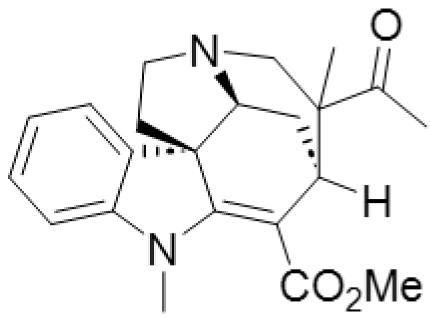
CC1(C(C)=O)[C@@]2([H])C(C(OC)=O)=C3NC4=CC=CC=C4[C@]3(CCN5C1)[C@@H]5C2

D.



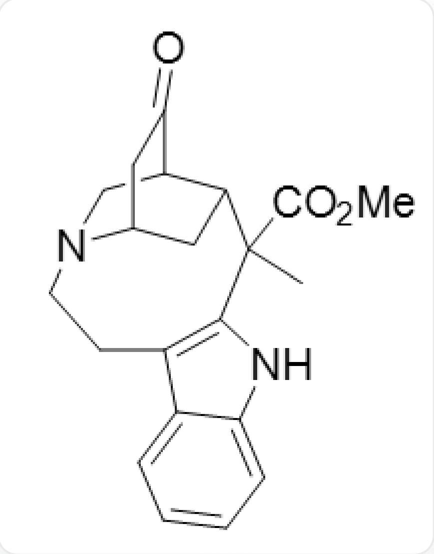
CC1(C(C)=O)[C@@]2([H])C(C(OC)=O)=C3N(C)C4=CC=CC=C4[C@]3(CC1N5C)[C@@H]5C2

E.



CC1(C(C)=O)[C@@]2([H])C(C(OC)=O)=C3N(C)C4=CC=CC=C4[C@]3(CCN5C1)[C@@H]5C2

F.



O=C1[C@@H]2C[N@@](CCC3=C4NC5=CC=CC=C35)[C@H](C1)C[C@H]2C4(C(OC)=O)C

Answer

Correct Answer: B

Detailed Explanation

The raw material first undergoes nitrogen methylation to obtain a quaternary ammonium salt, yielding intermediate **1**

CHECKPOINT 1 PTS

原料首先发生氮甲基化得到季铵盐得到中间体**1**

The structure of **1** is: CC([C@H]1C[N@@+]2(C)[C@H]3C[C@]1([H])C(C(OC)=O)=C4NC5=CC=CC=C5[C@@]43CC2)=O

CHECKPOINT 1 PTS

1 的结构为：CC([C@H]1C[N@@+]2(C)[C@H]3C[C@]1([H])C(C(OC)=O)=C4NC5=CC=CC=C5[C@@]43CC2)=O

Subsequently, intermediate **1** undergoes elimination under the action of NaOH to obtain **A**

CHECKPOINT 1 PTS

此后中间体 **1** 在 NaOH 的作用下消除得到 **A**

A: C=C(C(C)=O)[C@@]1([H])C(C(OC)=O)=C2NC3=CC=CC=C3[C@]2(CCN4C)[C@@H]4C1

CHECKPOINT 1 PTS

A: C=C(C(C)=O)[C@@]1([H])C(C(OC)=O)=C2NC3=CC=CC=C3[C@]2(CCN4C)[C@@H]4C1

A can undergo intramolecular hydride shift, experiencing intermediate **2**

CHECKPOINT 1 PTS

A 可以发生分子内的负氢迁移，经历中间体 **2**

Intermediate **2** undergoes intramolecular cyclization to obtain **B**:CC1(C(C)=O)[C@@]2([H])C(C(OC)=O)=C3NC4=CC=CC=C4[C@]3(CC1N5C)[C@@H]5C2。

CHECKPOINT 1 PTS

中间体 **2** 发生分子内关环得到 **B**:CC1(C(C)=O)[C@@]2([H])C(C(OC)=O)=C3NC4=CC=CC=C4[C@]3(CC1N5C)[C@@H]5C2。

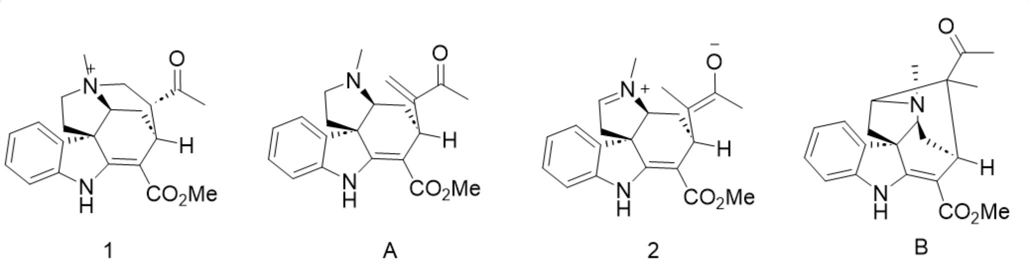
The structure of **2** :C/C([C@@]1([H])C(C(OC)=O)=C2NC3=CC=CC=C3[C@]2(CC=[N+]4C)[C@@H]4C1)=C([O-])/C

CHECKPOINT 1 PTS

2 :C/C([C@@]1([H])C(C(OC)=O)=C2NC3=CC=CC=C3[C@]2(CC=[N+]4C)[C@@H]4C1)=C([O-])/C

Because the BDE on the secondary carbon is smaller and it is also spatially closer, the hydrogen migration on the secondary carbon is selected.

Therefore, B is correct



1 : CC([C@H]1C[N@@+]2(C)[C@H]3C[C@]1([H])C(C(OC)=O)=C4NC5=CC=CC=C5[C@@]43CC2)=O A: C=C(C(C)=O)[C@@]1([H])C(C(OC)=O)=C2NC3=CC=CC=C3[C@]2(CCN4C)[C@@H]4C1 2 :C/C([C@@]1([H])C(C(OC)=O)=C2NC3=CC=CC=C3[C@]2(CC=[N+]4C)[C@@H]4C1)=C([O-])/C B: CC1(C(C)=O)[C@@]2([H])C(C(OC)=O)=C3NC4=CC=CC=C4[C@]3(CC1N5C)[C@@H]5C2